Practice workbook · Exponential equations, including hidden quadratics

Equations with the unknown in a power, solved by matching bases or taking logs, including those that become a quadratic or cubic in a substitution such as u = e^(3x) or u = 2^y.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-09

1. Question 11 · part a.ii · 3 marks

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Worked-example retry · Attempt part a.ii only.

11

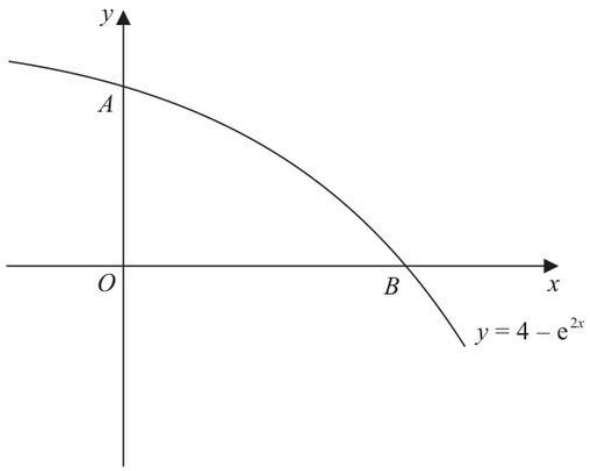


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = 4 - e^{2x}$
The curve C crosses the y-axis at the point A and the x-axis at the point B.

(a) (i) Write down the y coordinate of point A.

(ii) Show that the x coordinate of B is $x = \ln 2$

(3)

The line l is the normal to C at the point B.

(b) Find an equation for l, giving your answer in the form $y = mx + c$

(4)

The finite region R is bounded by C, l and the y-axis.

(c) Using calculus, find the area of R.
Give your answer to one decimal place.

(7)

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1. Question 11 · part a.ii · 3 marks · Working and answer

Working and answer

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2. Question 11 · part a.ii · 3 marks

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Further practice · Attempt part a.ii only.

11

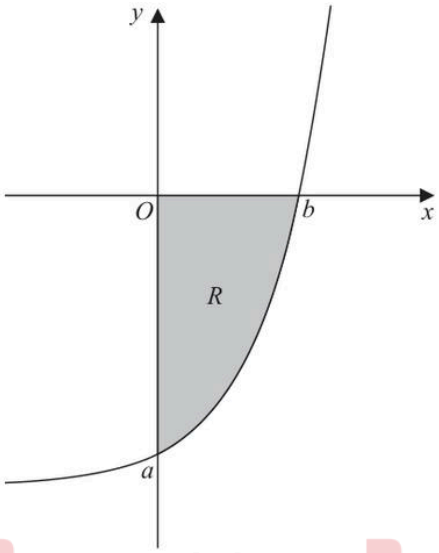


Diagram **NOT** accurately drawn

Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the curve with equation $y = e^{2x} - 9$ and the coordinate axes.

The curve crosses the coordinate axes at the points with coordinates $(0, a)$ and $(b, 0)$

(a) (i) Find the value of a

(ii) Show that $b = \ln 3$

(3)

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.

Give your answer in the form $\pi(p \ln 3 + q)$, where p and q are integers.

(6)

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2. Question 11 · part a.ii · 3 marks · Working and answer

Working and answer

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3. Question 10 · part a · 4 marks

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Further practice · Attempt part a only.

10 (a) Show that $\frac{9^{3y}}{243} = 3^{(6y-5)}$

(4)

(b) Solve the simultaneous equations

$$\frac{9^{3y}}{243} = 27^{(x-2)}$$

$$\log_{10}\sqrt{6xy} = \log_4 2$$

(9)

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3. Question 10 · part a · 4 marks · Working and answer

Working and answer

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4. Question 8 · part a · 4 marks

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Further practice · Attempt part a only.

8 The curve C_1 has equation $y = 5e^{-2x} + 4$

The curve C_2 has equation $y = e^{2x}$

The curves C_1 and C_2 intersect at the point A.

(a) Find the exact coordinates of A.

(4)

The tangent at A to C_1 intersects the x-axis at the point B.

(b) Show that the x coordinate of B is $\frac{1}{2}(5 + \ln 5)$

(5)

The tangent at A to C_2 intersects the x-axis at the point D.

(c) Find the area of ΔABD.

(6)

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4. Question 8 · part a · 4 marks · Working and answer

Working and answer

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5. Question 8 · part a.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

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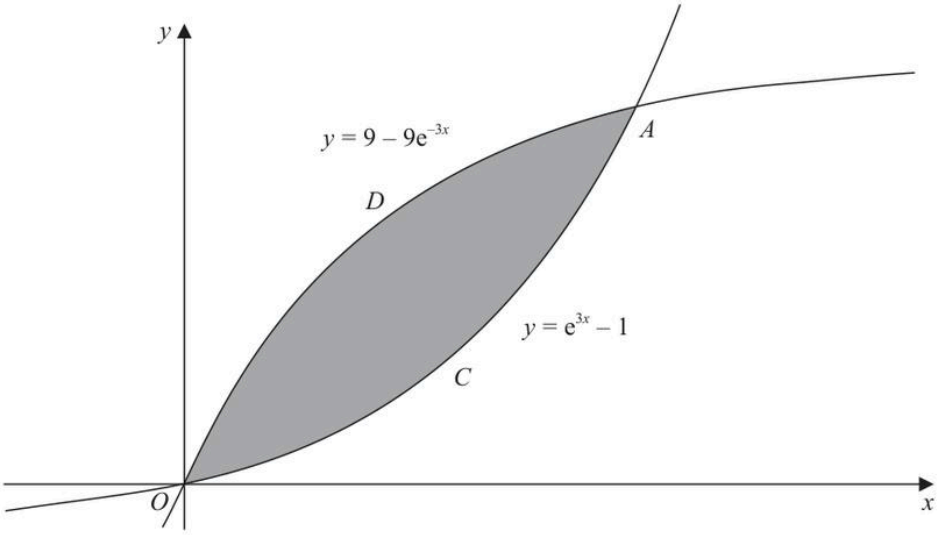


Figure 2

Figure 2 shows part of the curve C with equation $y = e^{3x} - 1$ and part of the curve D with equation $y = 9 - 9e^{-3x}$

The curves intersect at the origin O and the point A.

(a) (i) Show that the x coordinate of the point A satisfies the equation

$$(e^{3x})^2 - 10e^{3x} + 9 = 0$$

(ii) Hence, show that the x coordinate of the point A is $\frac{1}{3}\ln 9$

(5)

The finite region bounded by C and by D is shown shaded in Figure 2.

(b) Use calculus to find the exact area of this region.

(6)

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P 7 1 6 4 2 A 0 2 2 3 6

5. Question 8 · part a.i · 0 marks · Working and answer

Working and answer

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6. Question 8 · part a.ii · 5 marks

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Further practice · Attempt part a.ii only.

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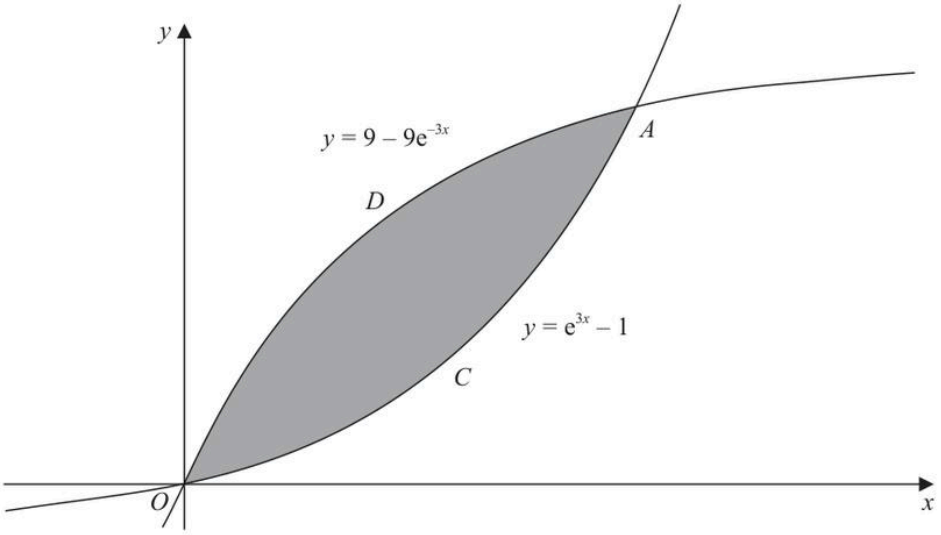


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(6)

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6. Question 8 · part a.ii · 5 marks · Working and answer

Working and answer

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7. Question 5 · part c · 5 marks

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Further practice · Attempt part c only.

5 $f(x) = 2ax^3 + x^2 - bx + 3a$ where a and b are integers.

Given that $(x + 2)$ and $(x - 1)$ are both factors of $f(x)$

(a) show that $a = 2$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(2)

Hence, given that $h(y) = 2^{(3y+2)} + 2^{2y} - 11(2^y) + 6$

(c) solve the equation $h(y) = 0$

Where appropriate give your answers to 3 decimal places.

(5)

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P 7 1 8 1 8 A 0 1 0 2 8

7. Question 5 · part c · 5 marks

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Further practice · Attempt part c only.



7. Question 5 · part c · 5 marks · Working and answer

Working and answer

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